INCH-POUND

MIL-R-6106/44A(USAF)
10 March 1989
SUPERSEDING
MIL-R-6106/44(USAF)
6 July 1984

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 50 AMPERES, 115/200 V AC, 400 HZ, 3PDT, SIDE STABLE, HERMETICALLY SEALED

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

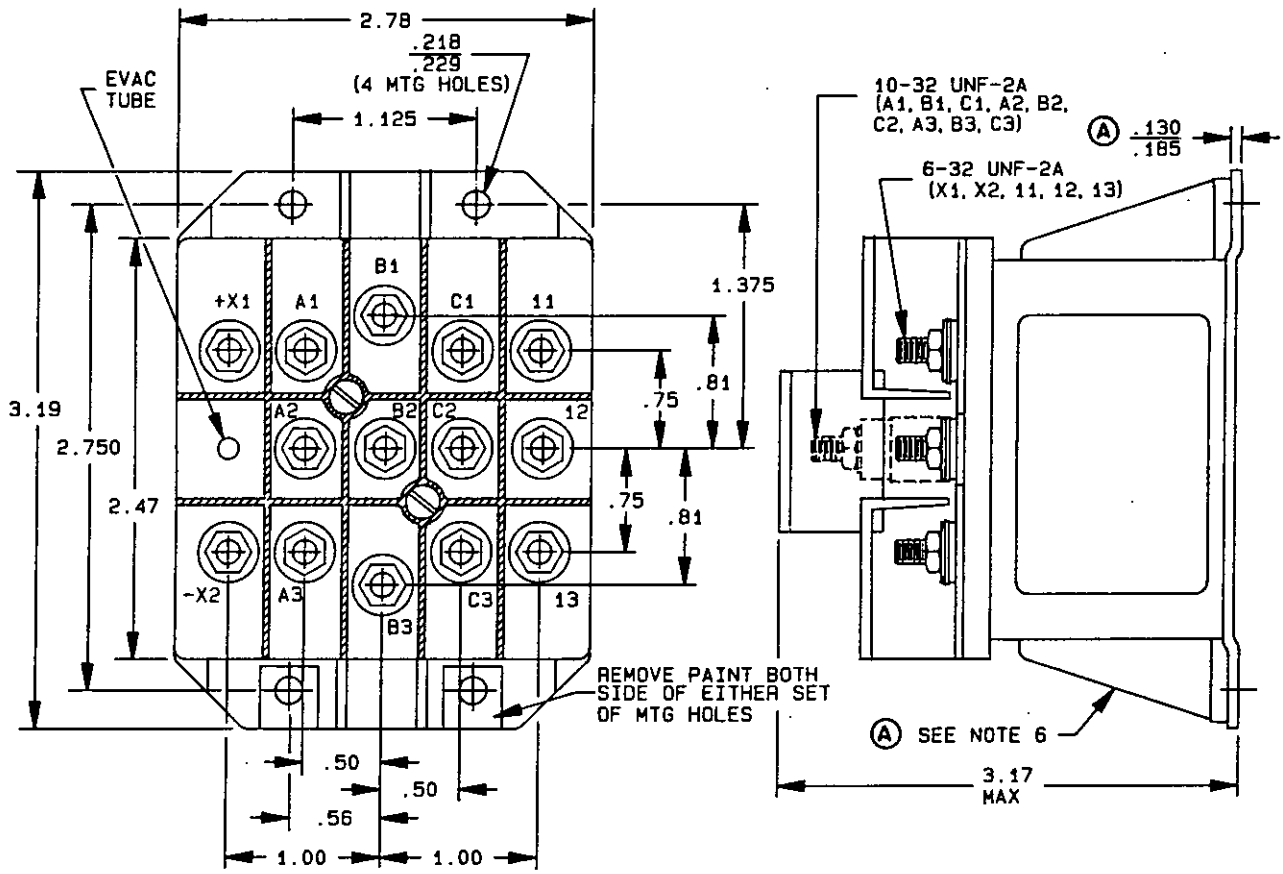
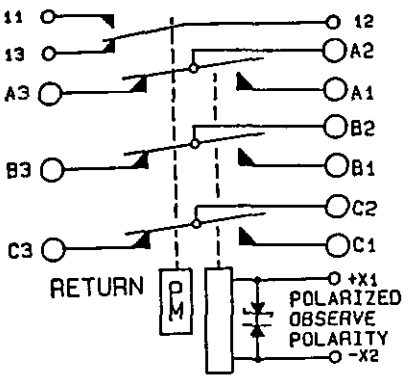


FIGURE 1. Relay, outline drawing.

Ⓐ denotes changes

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CIRCUIT DIAGRAM

Inches	mm
.130	3.30
.185	4.69
.218	5.54
.219	5.56
.229	5.82
.50	12.7
.56	14.2
.75	19.1
.81	20.6
1.00	25.4
1.125	28.58
1.375	34.93
2.47	62.7
2.750	69.85
2.78	70.6
3.17	80.52
3.19	81.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.030 (0.76 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
Ⓐ5. Intended use: The contactor specified herein shall be used in a high performance military aircraft electrical power system. It shall be used to turn off or transfer an electrical load from one power source of 115/200 V ac, 400 Hz, 3 phase power to an alternate source of the same type power. These powers will be applied to the normally open and normally closed contacts concurrently. The phase relation between the two sources will be random.
Ⓐ6. Support plates are optional.

FIGURE 1. Relay, outline drawing - Continued.

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REQUIREMENTS:

Coil data:

Duty cycle: Continuous.

Operating voltage:

Nominal: 28 V dc

Maximum: 30 V dc

Ⓐ Pickup voltage, maximum:

Ⓐ Normal: 18 V dc over the temperature range.

Ⓐ High temperature test: 19.5 V dc

Ⓐ Continous current test: 22.5 V dc.

Ⓐ Dropout voltage: 2.5 V dc over the temperature range.

Ⓐ Hold voltage: 6.5 V dc over the temperature range.

Current rating: 0.280 ampere maximum at +25°C.

Operating time:

Ⓐ Operate: 35 milliseconds maximum.

Ⓐ Release: 35 milliseconds maximum.

Ⓐ Transfer: 3 milliseconds maximum.

Bounce:

Main contacts: 2 milliseconds maximum.

Ⓐ Auxiliary contacts: 2 milliseconds maximum.

Coil resistance at +25°C: 100 ohms ±20 percent.

Coil transient voltage: ±50 volts.

Contact data:

Main contacts:

Configuration: 3PDT side stable.

Voltage nominal: 115/200 V ac 400 Hz.

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Current rating:

Resistive: 50 amperes.

Inductive: 50 amperes.

Motor: 50 amperes.

Lamp: 15 amperes.

Rupture: 500 amperes.

Overload: 500 amperes.

Contact transfer time: 1.8 milliseconds minimum.

Auxiliary contacts:

Configuration: SPDT.

Voltage: 28 V dc or 115 V ac 400 Hz.

Current rating:

Resistive: 5 amperes.

Inductive: 5 amperes.

Lamp: 1 ampere.

Life:

Electrical: 50,000 cycles minimum at rated load.

Mechanical: 100,000 cycles.

Physical data:

Strength of terminals: Applicable.

Base cover: Melamine fiberglass, MIL-P-15037.

Terminal barriers: Nylon 6/6, type 1, grade A, class 1, in accordance with LP-395, black.

Weight: 18 ounces maximum.

Finish: Instrument black.

Environmental data:

Ambient temperature range: -65°F to +160°F.

Vibration, sine: In accordance with MIL-E-5272, procedure XII, curve A.

Shock: 15 G maximum, 11 milliseconds.

Altitude: 60,000 feet maximum.

Part or Identifying Number (PIN): M6106/44-001.

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CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F687)

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